Elementary Analysis

Chapter 2: Sections 7–8

All 15 exercises, with every subpart
Detailed worked solutions and direct limit proofs

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Text and numbering. Kenneth A. Ross, *Elementary Analysis: The Theory of Calculus*, second edition (Springer, 2013), Chapter 2. Exercise numbers follow the copy already linked on Mathjock. These are independently written worked solutions, not the publisher's solutions manual. Each exercise is identified by a short mathematical description.

Reading philosophy. Every argument states its goal, records the assumptions being used, and explains the intermediate algebra or logical inference. Definitions, sign conditions, and both directions of an "if and only if" are made explicit. A cited rule is named at the point where it is used.

Conventions. $\mathbb{N} = \{1, 2, 3, \ldots\}$, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$, and $\mathbb{Q} = \{p/q : p, q \in \mathbb{Z},\ q \neq 0\}$. Unless a different domain is stated, sequence indices run through $\mathbb{N}$ and all terms and finite limits are real. The empty sum is 0, and $0! = 1$.

Coverage. Section 7: Exercises 7.1–7.5. Section 8: Exercises 8.1–8.10. These notes cover the opening two sections of Chapter 2.

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7 Limits of sequences

A sequence assigns a real number s_n to each index n. Unless the exercise states otherwise, the indices here are $n = 1, 2, 3, \ldots$. The order of the terms matters, even when their values repeat.

We use Ross's Definition 7.1: $s_n \to s$ means that, for every $\varepsilon > 0$, there is a number N such that

$$n > N \quad \Longrightarrow \quad |s_n - s| < \varepsilon \qquad (n \in \mathbb{N}).$$

The choice of N may depend on ε but cannot depend on the later index n. A real threshold N is permitted; the Archimedean property ensures that there are integer indices past it. Convergence here means convergence to a finite real number. Divergence means that no such real limit exists.

Although several exercises in this section ask only for answers, we include the algebra and estimates that explain those answers. The proofs use elementary inequalities and the limit definition, rather than the limit laws introduced in Section 9.

7.1 Exercise 7.1: The first five terms of four sequences

What we must show. Compute the terms with indices $n = 1, 2, 3, 4, 5$ for

$$\text{(a) } s_n = \frac{1}{3n+1}, \qquad \text{(b) } b_n = \frac{3n+1}{4n-1}, \qquad \text{(c) } c_n = \frac{n}{3^n}, \qquad \text{(d) } d_n = \sin\left(\frac{n\pi}{4}\right).$$

Plan. Substitute the index into every occurrence of n, evaluate the numerator and denominator separately, and simplify where possible. Indices start at 1. In part (a), the denominator $3n+1$ means 3 times n, followed by addition of 1.

1. Part (a): A reciprocal with a linear denominator

For $n \in \mathbb{N}$, we have $3n + 1 \geq 4 > 0$, so all the indicated fractions are defined. The five substitutions give

$$\begin{aligned}
s_1 &= \frac{1}{3 \cdot 1 + 1} = \frac{1}{3+1} = \frac{1}{4}, \\
s_2 &= \frac{1}{3 \cdot 2 + 1} = \frac{1}{6+1} = \frac{1}{7}, \\
s_3 &= \frac{1}{3 \cdot 3 + 1} = \frac{1}{9+1} = \frac{1}{10}, \\
s_4 &= \frac{1}{3 \cdot 4 + 1} = \frac{1}{12+1} = \frac{1}{13}, \\
s_5 &= \frac{1}{3 \cdot 5 + 1} = \frac{1}{15+1} = \frac{1}{16}.
\end{aligned}$$

2. Part (b): A quotient of two linear expressions

Here $4n - 1 \geq 3 > 0$ for every $n \geq 1$, so there is again no zero denominator. Compute both parts of each fraction:

$$b_1 = \frac{3 \cdot 1 + 1}{4 \cdot 1 - 1} = \frac{3+1}{4-1} = \frac{4}{3},$$

$$b_2 = \frac{3\cdot 2+1}{4\cdot 2-1} = \frac{6+1}{8-1} = \frac{7}{7} = 1,$$
$$b_3 = \frac{3\cdot 3+1}{4\cdot 3-1} = \frac{9+1}{12-1} = \frac{10}{11},$$
$$b_4 = \frac{3\cdot 4+1}{4\cdot 4-1} = \frac{12+1}{16-1} = \frac{13}{15},$$
$$b_5 = \frac{3\cdot 5+1}{4\cdot 5-1} = \frac{15+1}{20-1} = \frac{16}{19}.$$

3. Part (c): A linear numerator and an exponential denominator

The denominator is the positive integer power 3^n. In particular, $3^1 = 3$, $3^2 = 9$, $3^3 = 27$, $3^4 = 81$, and $3^5 = 243$. Substitution yields

$$c_1 = \frac{1}{3^1} = \frac{1}{3},$$
$$c_2 = \frac{2}{3^2} = \frac{2}{9},$$
$$c_3 = \frac{3}{3^3} = \frac{3}{27} = \frac{1}{9},$$
$$c_4 = \frac{4}{3^4} = \frac{4}{81},$$
$$c_5 = \frac{5}{3^5} = \frac{5}{243}.$$

The cancellation in the third term divides numerator and denominator by 3; it does not change the value of that term.

4. Part (d): Sine at multiples of an eighth-turn angle

On the unit circle at angle $\pi/4$, the two coordinates are equal and positive. If each is t, then $t^2+t^2 = 1$, so $t^2 = 1/2$ and $t = \sqrt{2}/2$. Thus $\sin(\pi/4) = \sqrt{2}/2$. The unit-circle values at $\pi/2$ and π are 1 and 0. Using these values and the identities $\sin(\pi-\theta) = \sin\theta$ and $\sin(\pi+\theta) = -\sin\theta$, we obtain

$$d_1 = \sin\left(\frac{1\pi}{4}\right) = \sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2},$$
$$d_2 = \sin\left(\frac{2\pi}{4}\right) = \sin\left(\frac{\pi}{2}\right) = 1,$$
$$d_3 = \sin\left(\frac{3\pi}{4}\right) = \sin\left(\pi - \frac{\pi}{4}\right) = \frac{\sqrt{2}}{2},$$
$$d_4 = \sin\left(\frac{4\pi}{4}\right) = \sin\pi = 0,$$
$$d_5 = \sin\left(\frac{5\pi}{4}\right) = \sin\left(\pi + \frac{\pi}{4}\right) = -\frac{\sqrt{2}}{2}.$$

Conclusion. The requested five-term lists are

Part	1	2	3	4	5
(a)	$\frac{1}{4}$	$\frac{1}{7}$	$\frac{1}{10}$	$\frac{1}{13}$	$\frac{1}{16}$
(b)	$\frac{4}{3}$	1	$\frac{10}{11}$	$\frac{13}{15}$	$\frac{16}{19}$
(c)	$\frac{1}{3}$	$\frac{2}{9}$	$\frac{1}{9}$	$\frac{4}{81}$	$\frac{5}{243}$
(d)	$\frac{\sqrt{2}}{2}$	1	$\frac{\sqrt{2}}{2}$	0	$-\frac{\sqrt{2}}{2}$

7.2 Exercise 7.2: Convergence of the four preceding sequences

What we must show. Decide which sequences in Exercise 7.1 converge and give each finite limit when it exists.

Plan. Use exact error estimates for the first three sequences. For the sine sequence, find values 1 and -1 at arbitrarily large indices, then compare them with any proposed common limit. The arguments use the limit definition directly: $u_n \to L$ means that for every $\varepsilon > 0$ there is a threshold N such that $n > N$ implies $|u_n - L| < \varepsilon$.

1. Part (a): The limit of $1/(3n+1)$

For every positive integer n,

$$\left|\frac{1}{3n+1} - 0\right| = \frac{1}{3n+1} < \frac{1}{3n}.$$

The reciprocal inequality follows because $3n+1 > 3n > 0$. Let $\varepsilon > 0$. By the Archimedean property, choose an integer $N > 1/(3\varepsilon)$. If $n > N$, then $3n > 1/\varepsilon$ and

$$\left|\frac{1}{3n+1} - 0\right| < \frac{1}{3n} < \varepsilon.$$

This verifies the definition and proves $s_n \to 0$.

2. Part (b): The limit of $(3n+1)/(4n-1)$

The ratio of the leading coefficients suggests $3/4$. Compute the error over a common denominator to verify it:

$$\begin{aligned} b_n - \frac{3}{4} &= \frac{3n+1}{4n-1} - \frac{3}{4} \\ &= \frac{4(3n+1) - 3(4n-1)}{4(4n-1)} \\ &= \frac{12n + 4 - 12n + 3}{4(4n-1)} \\ &= \frac{7}{4(4n-1)}. \end{aligned}$$

The denominator is positive for $n \geq 1$, so this is also the absolute error. Moreover, $4n - 1 - 3n = n - 1 \geq 0$, hence $4n - 1 \geq 3n$. Therefore

$$\left|b_n - \frac{3}{4}\right| = \frac{7}{4(4n-1)} \leq \frac{7}{12n}.$$

Given $\varepsilon > 0$, choose an integer $N > \max\{1, 7/(12\varepsilon)\}$. For $n > N$ we get

$$\left|b_n - \frac{3}{4}\right| \leq \frac{7}{12n} < \varepsilon.$$

Thus $b_n \to 3/4$. No quotient limit law is needed for this conclusion.

3. Part (c): Bound the exponential denominator by a square

We first prove $3^n \geq n^2$ for every integer $n \geq 2$. For $n = 2$, this reads $9 \geq 4$. Suppose it holds for an integer $n \geq 2$. Then

$$3^{n+1} = 3 \cdot 3^n \geq 3n^2.$$

Expanding the next square gives

$$3n^2 - (n+1)^2 = 3n^2 - (n^2 + 2n + 1) = 2n(n-1) - 1 \geq 3 > 0,$$

where $n \geq 2$ gives $2n(n-1) \geq 4$. Hence $3n^2 \geq (n+1)^2$, so $3^{n+1} \geq (n+1)^2$. Mathematical induction proves the claimed bound.

Because all denominators are positive, the bound implies, for $n \geq 2$,

$$0 < \frac{n}{3^n} \leq \frac{n}{n^2} = \frac{1}{n}.$$

Let $\varepsilon > 0$ and choose an integer $N > \max\{2, 1/\varepsilon\}$. Then $n > N$ gives

$$\left| \frac{n}{3^n} - 0 \right| = \frac{n}{3^n} \leq \frac{1}{n} < \varepsilon.$$

It follows directly that $c_n \to 0$.

4. Part (d): Values separated by 2 persist arbitrarily far out

For every nonnegative integer k, periodicity of sine gives

$$d_{8k+2} = \sin\left(\frac{(8k+2)\pi}{4}\right) = \sin\left(2k\pi + \frac{\pi}{2}\right) = 1,$$
$$d_{8k+6} = \sin\left(\frac{(8k+6)\pi}{4}\right) = \sin\left(2k\pi + \frac{3\pi}{2}\right) = -1.$$

Both families of indices can be made larger than any given real threshold N: choose an integer $k > \max\{0, N/8\}$. Then $8k + 2 > N$ and $8k + 6 > N$.

Suppose, for a contradiction, that $d_n \to L \in \mathbb{R}$. Use $\varepsilon = 1/2$ in the definition. There is a threshold N such that $n > N$ implies $|d_n - L| < 1/2$. Choose k so that both indices above exceed this N. Then

$$|1 - L| < \frac{1}{2}, \qquad |-1 - L| < \frac{1}{2}.$$

The triangle inequality, together with $|L + 1| = |-1 - L|$, would imply

$$2 = |1 - (-1)| = |(1 - L) + (L + 1)| \leq |1 - L| + |L + 1| < \frac{1}{2} + \frac{1}{2} = 1.$$

This contradicts $2 > 1$. Therefore the sine sequence has no real limit.

Conclusion. Parts (a), (b), and (c) converge to 0, 3/4, and 0, respectively. Part (d) diverges.

7.3 Exercise 7.3: A catalogue of twenty sequences and their limits

What we must show. Determine convergence or divergence for every sequence in parts (a)–(t), and give the real limit whenever one exists. Although the exercise does not require proofs, we include a justification of every answer. The arguments use the definition of convergence directly, rather than the limit laws developed later in Section 9.

Plan. For a convergent sequence, subtract the proposed limit and bound the resulting error by a reciprocal that becomes arbitrarily small. For a divergent sequence, find either arbitrarily late values that are arbitrarily large, or two arbitrarily late values separated by a fixed positive distance.

Elementary facts used in the individual parts

Fact 1: Reciprocal error estimates imply convergence.

The definition says that $u_n \to L$ if, for every $\varepsilon > 0$, there is a threshold N such that

$$n > N \quad \Longrightarrow \quad |u_n - L| < \varepsilon.$$

Suppose we prove $|u_n - L| \le C/n$ for every $n \ge n_0$, where $C > 0$ is fixed. Given $\varepsilon > 0$, choose a positive integer $N > \max\{n_0, C/\varepsilon\}$, using the Archimedean property. Then $n > N$ implies both $n \ge n_0$ and $n > C/\varepsilon$. Multiplying the latter inequality by $\varepsilon > 0$ and dividing by $n > 0$ gives

$$|u_n - L| \le \frac{C}{n} < \varepsilon.$$

This proves convergence from the definition. In particular, $1/n \to 0$. If an error is identically 0, it is less than every $\varepsilon > 0$ without any additional estimate.

We will also use a shifted version. If, for all sufficiently large n, $|u_n - L| \le C/(n - r)$, with fixed real r and $n > r$, choose N larger than the starting index and $r + C/\varepsilon$. Then $n > N$ gives $n - r > C/\varepsilon$, and the same calculation gives $|u_n - L| < \varepsilon$. This proves the shifted version without any new limit theorem.

Only the tail is used in either proof. Thus a finite number of unusual initial values does not affect the conclusion: increase N to exceed all their indices, and every $n > N$ still satisfies the required estimate. This observation will matter when an initial denominator is negative but never zero.

Fact 2: Powers of two dominate a simple linear expression.

For every integer $m \ge 0$,

$$2^m \ge m + 1.$$

At $m = 0$, both sides equal 1. If the inequality holds at m, then multiplication by $2 > 0$ gives

$$2^{m+1} = 2 \cdot 2^m \ge 2(m + 1) = 2m + 2 \ge m + 2,$$

where the last difference is $(2m + 2) - (m + 2) = m \ge 0$. Induction proves the assertion. Taking reciprocals of positive quantities consequently gives

$$0 < 2^{-m} \le \frac{1}{m + 1}.$$

Fact 3: Separated arbitrarily late values rule out a real limit.

Suppose some fixed $D > 0$ has this property: for every threshold N, there are indices $p, q > N$ with $|u_p - u_q| \ge D$. Assume, to obtain a contradiction, that $u_n \to L \in \mathbb{R}$. Apply the definition with $\varepsilon = D/3$. It supplies N such that both late terms obey $|u_p - L| < D/3$ and $|u_q - L| < D/3$. The triangle inequality would then give

$$|u_p - u_q| = |(u_p - L) + (L - u_q)| \le |u_p - L| + |L - u_q| < \frac{D}{3} + \frac{D}{3} = \frac{2D}{3} < D,$$

contradicting the stipulated separation. Thus such a sequence does not converge to a real number. This argument does not assume any theorem about subsequences.

Fact 4: Arbitrarily large late values also rule out a real limit.

Suppose that for every $B \in \mathbb{R}$ and every threshold N, some $n > N$ satisfies $u_n > B$. If $u_n \to L \in \mathbb{R}$, the definition with $\varepsilon = 1$ would supply N for which $|u_n - L| < 1$ whenever $n > N$. Choose a late value with $u_n > L + 1$. It satisfies $u_n - L > 1$, and therefore $|u_n - L| = u_n - L > 1$, a contradiction. This is a direct nonconvergence argument; it does not invoke a later boundedness theorem for convergent sequences.

Solutions of parts (a)–(t)

(a) The sequence $a_n = n/(n+1)$.

For every $n \in \mathbb{N}$, $n + 1 \geq 2 > 0$, so every term is defined. To test the candidate 1, subtract it using the common denominator $n + 1$:

$$a_n - 1 = \frac{n}{n+1} - \frac{n+1}{n+1} = \frac{n - (n+1)}{n+1} = -\frac{1}{n+1}.$$

It follows that

$$|a_n - 1| = \frac{1}{n+1} \leq \frac{1}{n}.$$

The inequality uses $n + 1 \geq n > 0$ and the reversal of order for positive reciprocals. Fact 1 with $C = 1$ now proves the limit.

Conclusion. Part (a) converges, and $\lim_{n\to\infty} \dfrac{n}{n+1} = 1$.

(b) The sequence $b_n = (n^2 + 3)/(n^2 - 3)$.

The denominator is nonzero at every positive integer. At $n = 1$ it is $1 - 3 = -2$, while $n \geq 2$ implies $n^2 \geq 4$ and $n^2 - 3 \geq 1 > 0$. Thus there is no undefined term, including in the initial part of the sequence.

Subtracting the candidate 1 gives

$$b_n - 1 = \frac{n^2 + 3 - (n^2 - 3)}{n^2 - 3} = \frac{n^2 + 3 - n^2 + 3}{n^2 - 3} = \frac{6}{n^2 - 3}.$$

For $n \geq 3$, we have $n^2 \geq 9 \geq 6$, so $3 \leq n^2/2$. Subtracting 3 from n^2 therefore gives

$$n^2 - 3 \geq n^2 - \frac{n^2}{2} = \frac{n^2}{2} > 0.$$

Consequently

$$|b_n - 1| = \frac{6}{n^2 - 3} \leq \frac{6}{n^2/2} = \frac{12}{n^2} \leq \frac{12}{n}.$$

The last inequality follows from $n^2 \geq n > 0$ for $n \geq 1$. Fact 1 with $C = 12$ and starting index 3 proves convergence. The negative first denominator does not enter the tail estimate.

Conclusion. Part (b) converges to 1.

(c) The sequence $c_n = 2^{-n}$.

The expression means the positive reciprocal $1/2^n$. Fact 2 gives

$$|c_n - 0| = \frac{1}{2^n} \leq \frac{1}{n+1} \leq \frac{1}{n} \qquad (n \geq 1).$$

The error is thus bounded by $1/n$. Applying Fact 1 proves the limit directly.

Conclusion. Part (c) converges to 0.

(d) The sequence $t_n = 1 + 2/n$.

Since $n > 0$, every term is defined, and subtracting 1 gives

$$|t_n - 1| = \left|1 + \frac{2}{n} - 1\right| = \left|\frac{2}{n}\right| = \frac{2}{n}.$$

For each $\varepsilon > 0$, choosing $N > 2/\varepsilon$ makes this error smaller than ε whenever $n > N$. This is Fact 1 with $C = 2$.

Conclusion. Part (d) converges to 1.

(e) The sequence $x_n = 73 + (-1)^n$.

If n is even, $(-1)^n = 1$ and $x_n = 73 + 1 = 74$. If n is odd, $(-1)^n = -1$ and $x_n = 73 - 1 = 72$. These values do not merely occur at the start: for any threshold N, choose a positive integer $m > N$. Then $2m > N$ and $2m + 1 > N$, and

$$x_{2m} = 74, \qquad x_{2m+1} = 72, \qquad |x_{2m} - x_{2m+1}| = |74 - 72| = 2.$$

Thus Fact 3 applies with $D = 2$. No real number can be approached by every term, because every tail still contains these two values at distance 2.

Conclusion. Part (e) does not converge.

(f) The sequence $s_n = 2^{1/n}$.

Write $r_n = 2^{1/n}$ for the positive real nth root of 2; thus $r_n > 0$ and $r_n^n = 2$. We prove the concrete estimate

$$0 \le r_n - 1 \le \frac{1}{n}.$$

It will imply convergence by Fact 1 without any theorem about limits of roots or logarithms.

First prove the auxiliary Bernoulli inequality. For every $x \ge 0$ and every positive integer m,

$$(1 + x)^m \ge 1 + mx.$$

Fix $x \ge 0$. At $m = 1$ the two sides are equal. If the inequality holds at m, multiplying by the positive number $1 + x$ gives

$$\begin{aligned}(1 + x)^{m+1} &\ge (1 + mx)(1 + x)\\ &= 1 + x + mx + mx^2\\ &= 1 + (m + 1)x + mx^2\\ &\ge 1 + (m + 1)x,\end{aligned}$$

because $mx^2 \ge 0$. This is the required step at $m + 1$, so induction proves the inequality for the fixed x, and hence for every $x \ge 0$.

Apply it with $x = 1/n$ and $m = n$. We obtain

$$\left(1 + \frac{1}{n}\right)^n \ge 1 + n\frac{1}{n} = 2.$$

For positive real numbers, raising to a positive integer power preserves strict order. For example, if $u > v > 0$, then $u/v > 1$, so the product of n factors u/v is greater than 1; multiplication by $v^n > 0$ gives $u^n > v^n$. If $r_n > 1 + 1/n$, this order property would give

$$2 = r_n^n > \left(1 + \frac{1}{n}\right)^n \geq 2,$$

a contradiction. Hence $r_n \leq 1 + 1/n$. Also $r_n \geq 1$: if $0 < r_n < 1$, multiplying n factors each less than 1 would give $r_n^n < 1$, contradicting $r_n^n = 2$.

Thus $1 \leq r_n \leq 1 + 1/n$, and subtraction of 1 yields

$$|r_n - 1| = r_n - 1 \leq \frac{1}{n}.$$

Fact 1 proves the asserted limit.

Conclusion. Part (f) converges to 1.

(g) The sequence $y_n = n!$.

For $n \geq 1$, the factorial can be written

$$n! = n\,(1 \cdot 2 \cdots (n-1)).$$

At $n = 1$, the parenthesized product is empty and equals 1. For $n \geq 2$, every factor in that product is at least 1, so its product is at least 1. In all cases, therefore,

$$n! \geq n.$$

Now let a real number B and a threshold N be arbitrary. Choose $n \in \mathbb{N}$ with $n > \max\{N, B, 1\}$. Then $n > N$ and $y_n = n! \geq n > B$. Thus arbitrarily late values exceed every prescribed real size. Fact 4 rules out convergence to any real number. This argument establishes the needed nonconvergence without citing a boundedness theorem.

Conclusion. Part (g) does not converge to a real limit; its values grow beyond every real bound.

(h) The sequence $d_n = (-1)^n n$.

At an even index $2m$ the value is $2m$, while at an odd index $2m + 1$ the value is $-(2m + 1)$. Given any threshold N, choose a positive integer $m > N$. Both indices $2m$ and $2m + 1$ are then greater than N, and

$$\begin{aligned} |d_{2m} - d_{2m+1}| &= |2m - (-(2m+1))| \\ &= |4m + 1| \\ &= 4m + 1 \geq 5 > 1. \end{aligned}$$

Here $m \geq 1$, so $4m + 1$ is positive, justifying the removal of the absolute value. Every tail has two values separated by at least 1. Fact 3 with $D = 1$ rules out a real limit.

Conclusion. Part (h) does not converge. The signs alternate while the magnitudes increase.

(i) The sequence $u_n = (-1)^n/n$.

Although its sign alternates, its distance from 0 is

$$|u_n - 0| = \frac{|(-1)^n|}{|n|} = \frac{1}{n},$$

because $|(-1)^n| = 1$ and $n > 0$. Fact 1 applies with $C = 1$. In contrast with (h), the alternating values now have small magnitude for every sufficiently large index.

Conclusion. Part (i) converges to 0.

(j) The sequence $u_n = (7n^3 + 8n)/(2n^3 - 3)$.

At $n = 1$, the denominator is $2 - 3 = -1$, so it is nonzero. For $n \geq 2$, $n^3 \geq 8$, giving $2n^3 - 3 \geq 16 - 3 = 13 > 0$. Thus the sequence is defined at every positive integer.

We test the candidate $7/2$ by exact subtraction:

$$\begin{aligned} u_n - \frac{7}{2} &= \frac{2(7n^3 + 8n) - 7(2n^3 - 3)}{2(2n^3 - 3)} \\ &= \frac{14n^3 + 16n - 14n^3 + 21}{2(2n^3 - 3)} \\ &= \frac{16n + 21}{2(2n^3 - 3)}. \end{aligned}$$

For $n \geq 2$, we have $n^3 \geq 8 > 3$, so

$$2n^3 - 3 \geq n^3 > 0.$$

Also $n \geq 1$ implies $21 \leq 21n$ and hence $16n + 21 \leq 37n$. Using these positive denominator bounds gives

$$\left| u_n - \frac{7}{2} \right| = \frac{16n + 21}{2(2n^3 - 3)} \leq \frac{37n}{2n^3} = \frac{37}{2n^2} \leq \frac{37}{2n}.$$

The last step again uses $n^2 \geq n > 0$. Fact 1, with $C = 37/2$ and starting index 2, proves convergence.

Conclusion. Part (j) converges to $7/2$.

(k) The sequence $u_n = (9n^2 - 18)/(6n + 18)$.

The denominator $6n + 18$ is positive for every $n \geq 1$. We establish a lower estimate that becomes arbitrarily large. For $n \geq 3$, $n^2 \geq 9$, so $3n^2 \geq 27 > 18$ and

$$9n^2 - 18 \geq 9n^2 - 3n^2 = 6n^2 > 0.$$

Also $n \geq 3$ implies $18 \leq 6n$, giving

$$0 < 6n + 18 \leq 12n.$$

First lower the numerator while keeping the positive denominator fixed, then increase that denominator:

$$u_n = \frac{9n^2 - 18}{6n + 18} \geq \frac{6n^2}{6n + 18} \geq \frac{6n^2}{12n} = \frac{n}{2}.$$

The second comparison follows from order reversal for positive reciprocals, followed by multiplication by $6n^2 > 0$.

Let $B \in \mathbb{R}$ and a threshold N be arbitrary. Choose a positive integer $n > \max\{N, 3, 2B\}$. The estimate applies and gives $u_n \geq n/2 > B$, with $n > N$. Fact 4 therefore rules out convergence to every real candidate.

Conclusion. Part (k) does not converge to a real limit; its values grow beyond every real bound.

(l) The sequence $u_n = \sin(n\pi/2)$.

Every positive integer has the form $n = 4q + r$, with $q \geq 0$ an integer and $r \in \{1, 2, 3, 4\}$. Since $n\pi/2 = 2q\pi + r\pi/2$, the period 2π of sine gives the repeating values

r	1	2	3	4
$\sin(r\pi/2)$	1	0	-1	0

The repetition alone suggests nonconvergence; we now check the arbitrarily late separation required by Fact 3. Given a threshold N, choose $m \in \mathbb{N}$ with $m > N$. Then $p = 4m+1 > N$ and $q = 4m+3 > N$, and periodicity gives

$$u_p = \sin(2m\pi + \pi/2) = 1, \qquad u_q = \sin(2m\pi + 3\pi/2) = -1.$$

Their distance is $|1 - (-1)| = 2$. Fact 3 applies with $D = 2$ and proves nonconvergence.

Conclusion. Part (l) does not converge.

(m) The sequence $u_n = \sin(n\pi)$.

For every integer n, the angle $n\pi$ lies at an integer multiple of a half-turn, so its sine is 0. Explicitly, if $n = 2m$ is even, then $\sin(n\pi) = \sin(2m\pi) = 0$ by periodicity. If $n = 2m + 1$ is odd, then

$$\sin(n\pi) = \sin(2m\pi + \pi) = \sin \pi = 0.$$

These two cases cover every positive integer. Thus $u_n = 0$ for every index, and for every $\varepsilon > 0$,

$$|u_n - 0| = 0 < \varepsilon$$

holds at every index. This verifies convergence directly from the definition.

Conclusion. Part (m) converges to 0.

(n) The sequence $u_n = \sin(2n\pi/3)$.

Write $n = 3q + r$, with $q \geq 0$ and $r \in \{1, 2, 3\}$. Then $2n\pi/3 = 2q\pi + 2r\pi/3$, giving the period table

r	1	2	3
$\sin(2r\pi/3)$	$\sqrt{3}/2$	$-\sqrt{3}/2$	0

For any threshold N, choose a positive integer $m > N$. The indices $p = 3m + 1$ and $q = 3m + 2$ both exceed N, and

$$u_p = \frac{\sqrt{3}}{2}, \qquad u_q = -\frac{\sqrt{3}}{2}.$$

Their distance is

$$|u_p - u_q| = \left| \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} \right| = \sqrt{3} > 0.$$

Consequently Fact 3 with $D = \sqrt{3}$ excludes every real limit.

Conclusion. Part (n) does not converge.

(o) The sequence $u_n = (\sin n)/n$.

Here the argument is n radians, with no factor of π. We do not need to classify the behavior of $\sin n$ itself. The elementary trigonometric inequality $|\sin x| \leq 1$ for every real x gives

$$|u_n - 0| = \left| \frac{\sin n}{n} \right| = \frac{|\sin n|}{n} \leq \frac{1}{n},$$

where division by $n > 0$ preserves the inequality. Fact 1 with $C = 1$ proves the limit.

Conclusion. Part (o) converges to 0.

(p) The sequence $u_n = (2^{n+1} + 5)/(2^n - 7)$.

The denominator is $2^n - 7$. At $n = 1$ it is $2 - 7 = -5$, and at $n = 2$ it is $4 - 7 = -3$. For $n \geq 3$, $2^n \geq 8$ and therefore $2^n - 7 \geq 1 > 0$. These cases show that every term is defined. For example the first three terms are

$$u_1 = \frac{4+5}{2-7} = -\frac{9}{5}, \qquad u_2 = \frac{8+5}{4-7} = -\frac{13}{3}, \qquad u_3 = \frac{16+5}{8-7} = 21.$$

They do not by themselves determine the limiting behavior.

Subtract the candidate 2 exactly:

$$\begin{aligned} u_n - 2 &= \frac{2^{n+1} + 5 - 2(2^n - 7)}{2^n - 7} \\ &= \frac{2 \cdot 2^n + 5 - 2 \cdot 2^n + 14}{2^n - 7} \\ &= \frac{19}{2^n - 7}. \end{aligned}$$

For $n \geq 4$, $2^n \geq 16 \geq 14$, so $7 \leq 2^n/2$. It follows that

$$2^n - 7 \geq 2^n - \frac{2^n}{2} = \frac{2^n}{2} > 0.$$

Using this bound and Fact 2 yields

$$\begin{aligned} |u_n - 2| &= \frac{19}{2^n - 7} \\ &\leq \frac{19}{2^n/2} = \frac{38}{2^n} \leq \frac{38}{n+1} \leq \frac{38}{n}. \end{aligned}$$

Fact 1 with $C = 38$ and starting index 4 proves convergence. It also explains why the negative initial terms and the relatively large value $u_3 = 21$ do not change the tail conclusion.

Conclusion. Part (p) converges to 2.

(q) The sequence $a_n = 3^n/n!$.

Both 3^n and $n!$ are positive for every $n \in \mathbb{N}$, so $a_n > 0$ and every ratio below is defined. Compare successive terms using $(n+1)! = (n+1)n!$:

$$\begin{aligned} \frac{a_{n+1}}{a_n} &= \frac{3^{n+1}}{(n+1)!} \cdot \frac{n!}{3^n} \\ &= \frac{3 \cdot 3^n \cdot n!}{(n+1)n!\, 3^n} \\ &= \frac{3}{n+1}. \end{aligned}$$

For $n \geq 5$, $n + 1 \geq 6$, giving

$$0 < \frac{3}{n+1} \leq \frac{3}{6} = \frac{1}{2}, \qquad a_{n+1} \leq \frac{1}{2} a_n.$$

We will turn this ratio estimate into an explicit reciprocal bound; no theorem about ratios of sequences is being invoked.

The value at the starting index is

$$C = a_5 = \frac{3^5}{5!} = \frac{243}{120} = \frac{81}{40} > 0.$$

We claim, for every integer $m \geq 0$, that

$$a_{5+m} \leq C\,2^{-m}.$$

At $m = 0$, this is equality $a_5 = C$. If the estimate holds at m, the ratio bound applies at $n = 5+m \geq 5$, and gives

$$a_{5+(m+1)} = a_{6+m} \leq \frac{1}{2}a_{5+m} \leq \frac{1}{2}C\,2^{-m} = C\,2^{-(m+1)}.$$

This is the next estimate, so induction proves the claim. Putting $m = n - 5$ now yields, for every $n \geq 5$,

$$0 < a_n \leq C\,2^{-(n-5)}.$$

Fact 2, applied to $m = n - 5 \geq 0$, supplies

$$2^{n-5} \geq (n-5) + 1 = n - 4 > 0.$$

Taking reciprocals and multiplying by $C > 0$ therefore gives

$$|a_n - 0| = a_n \leq C\,2^{-(n-5)} \leq \frac{C}{n-4} = \frac{81/40}{n-4}.$$

For any $\varepsilon > 0$, choose an integer $N > \max\{5, 4 + C/\varepsilon\}$. If $n > N$, then $n - 4 > C/\varepsilon$, so the displayed error is less than ε. This is the shifted estimate of Fact 1, proving the limit directly.

Conclusion. Part (q) converges to 0.

(r) The sequence $u_n = (1 + 1/n)^2$.

Expand by multiplying its two factors, not by assuming a rule about limits of squares:

$$\begin{aligned} u_n &= \left(1 + \frac{1}{n}\right)\left(1 + \frac{1}{n}\right) \\ &= 1 + \frac{1}{n} + \frac{1}{n} + \frac{1}{n^2} \\ &= 1 + \frac{2}{n} + \frac{1}{n^2}. \end{aligned}$$

All terms after the leading 1 are positive. Since $n^2 \geq n > 0$, we have $1/n^2 \leq 1/n$, giving

$$|u_n - 1| = \frac{2}{n} + \frac{1}{n^2} \leq \frac{2}{n} + \frac{1}{n} = \frac{3}{n}.$$

Fact 1 with $C = 3$ proves convergence.

Conclusion. Part (r) converges to 1.

(s) The sequence $u_n = (4n^2 + 3)/(3n^2 - 2)$.

For all $n \geq 1$, $n^2 \geq 1$, so $3n^2 - 2 \geq 3 - 2 = 1 > 0$. Thus every term is defined and the denominator has a fixed positive sign. Subtracting $4/3$ with a common denominator gives

$$\begin{aligned} u_n - \frac{4}{3} &= \frac{3(4n^2+3) - 4(3n^2-2)}{3(3n^2-2)} \\ &= \frac{12n^2 + 9 - 12n^2 + 8}{3(3n^2-2)} \\ &= \frac{17}{3(3n^2-2)}. \end{aligned}$$

The estimate $n^2 \geq 1$ also yields

$$(3n^2 - 2) - n^2 = 2n^2 - 2 = 2(n^2 - 1) \geq 0,$$

so $3n^2 - 2 \geq n^2 > 0$. Therefore

$$\left| u_n - \frac{4}{3} \right| = \frac{17}{3(3n^2-2)} \leq \frac{17}{3n^2} \leq \frac{17}{3n}.$$

Fact 1 with $C = 17/3$ verifies the limit from the definition.

Conclusion. Part (s) converges to $4/3$.

(t) The sequence $u_n = (6n + 4)/(9n^2 + 7)$.

For $n \geq 1$, both $6n + 4$ and $9n^2 + 7$ are positive, so the absolute value of u_n is u_n itself. We have

$$9n^2 + 7 \geq 9n^2 > 0, \qquad 6n + 4 \leq 6n + 4n = 10n,$$

where $4 \leq 4n$ uses $n \geq 1$. Taking reciprocals in the denominator comparison and then applying the numerator comparison,

$$|u_n - 0| = \frac{6n+4}{9n^2+7} \leq \frac{6n+4}{9n^2} \leq \frac{10n}{9n^2} = \frac{10}{9n}.$$

Fact 1 with $C = 10/9$ proves convergence to the proposed value 0.

Conclusion. Part (t) converges to 0.

Complete answer list

In this list, "does not converge" means that there is no finite real limit. Every entry is justified in its corresponding part above. The growth cases (g) and (k), the persistent oscillations (e), (h), (l), and (n), and the shrinking oscillations (i) and (o) have been distinguished by direct estimates and the definition.

Part	Answer	Part	Answer
(a)	Converges to 1	(k)	Does not converge
(b)	Converges to 1	(l)	Does not converge
(c)	Converges to 0	(m)	Converges to 0
(d)	Converges to 1	(n)	Does not converge
(e)	Does not converge	(o)	Converges to 0
(f)	Converges to 1	(p)	Converges to 2
(g)	Does not converge	(q)	Converges to 0
(h)	Does not converge	(r)	Converges to 1
(i)	Converges to 0	(s)	Converges to 4/3
(j)	Converges to 7/2	(t)	Converges to 0

7.4 Exercise 7.4: Rational and irrational terms can have different kinds of limits

What we must show. Give (a) a sequence of irrational numbers converging to a rational number, and (b) a sequence of rational numbers converging to an irrational number.

Plan. For part (a), divide $\sqrt{2}$ by larger positive integers. For part (b), use the lower decimal truncations of $\sqrt{2}$. We verify both the nature of every term and convergence.

1. Recall why $\sqrt{2}$ is irrational

The irrationality of $\sqrt{2}$ was established in Section 2. Here is the short integer argument as well. If $\sqrt{2} = p/q$ in lowest terms, with integers p, q and $q > 0$, then $p^2 = 2q^2$, so p^2 is even. The square of an odd integer $2j + 1$ is $4j(j + 1) + 1$, which is odd. Thus p must be even; write $p = 2j$. Substitution gives

$$4j^2 = 2q^2, \qquad q^2 = 2j^2.$$

Now q^2 is even, so the same odd-square argument makes q even. The numerator and denominator would have the common factor 2, contrary to lowest terms. Hence $\sqrt{2} \notin \mathbb{Q}$.

2. Part (a): Irrational terms with rational limit

Define, for every $n \in \mathbb{N}$,

$$x_n = \frac{\sqrt{2}}{n}.$$

Since $n \geq 1$, division is legitimate. If any term x_n were rational, multiplication by the rational integer n would make $nx_n = \sqrt{2}$ rational. This contradicts Step 1. Thus every term x_n is irrational.

To prove convergence, let $\varepsilon > 0$ and choose an integer $N > \sqrt{2}/\varepsilon$ by the Archimedean property. For $n > N$,

$$|x_n - 0| = \frac{\sqrt{2}}{n} < \varepsilon.$$

Therefore $x_n \to 0$. The limit is rational because $0 = 0/1$.

3. Part (b): Define each decimal truncation using a finite maximum

For a real $t \geq 0$, the notation $\lfloor t \rfloor$ denotes the integer m satisfying

$$m \leq t < m + 1.$$

To justify existence, choose a positive integer $M > t$ using the Archimedean property. The finite set

$$F = \{j \in \mathbb{Z} : 0 \leq j \leq M \text{ and } j \leq t\}$$

is nonempty because it contains 0. Let $m = \max F$, which exists for a nonempty finite set of integers. Then $m \leq t$. If $m + 1 \leq t$, the inequalities $0 \leq m + 1 \leq t < M$ would give $m + 1 \in F$, contradicting the maximality of m. Hence $t < m + 1$, as required.

This integer is unique. If two integers $m < j$ satisfied their respective inequalities, then $m + 1 \leq j \leq t$ would contradict $t < m + 1$. Interchanging them handles the other possible order. Thus the floor notation is well-defined here.

Apply this construction to $t = 10^n\sqrt{2}$ and put

$$m_n = \left\lfloor 10^n\sqrt{2} \right\rfloor, \qquad r_n = \frac{m_n}{10^n} \quad (n \in \mathbb{N}).$$

Every m_n is an integer and 10^n is a positive integer. Therefore every r_n is rational by the definition of $\mathbb{Q}$. The defining floor inequalities give

$$m_n \leq 10^n\sqrt{2} < m_n + 1.$$

Dividing by $10^n > 0$ preserves order and yields

$$r_n \leq \sqrt{2} < r_n + \frac{1}{10^n}.$$

After subtracting r_n, this becomes the precise truncation error estimate

$$0 \leq \sqrt{2} - r_n < \frac{1}{10^n}.$$

In fact the left inequality is strict, because a rational r_n cannot equal the irrational number $\sqrt{2}$.

4. Prove convergence of the rational truncations

We may compare the error with $1/n$. The inequality $10^n \geq n$ holds for all positive integers n: the base case is $10 \geq 1$; if it holds for n, then

$$10^{n+1} = 10 \cdot 10^n \geq 10n \geq n + 1,$$

because $10n - (n + 1) = 9n - 1 \geq 8 > 0$ for $n \geq 1$. Induction proves the inequality. Taking reciprocals of positive quantities gives $1/10^n \leq 1/n$.

Let $\varepsilon > 0$ and choose an integer $N > 1/\varepsilon$. For $n > N$, we obtain

$$\left|r_n - \sqrt{2}\right| = \sqrt{2} - r_n < \frac{1}{10^n} \leq \frac{1}{n} < \varepsilon.$$

Thus $r_n \to \sqrt{2}$, and its limit is irrational by Step 1.

Conclusion. For (a), take $x_n = \sqrt{2}/n \to 0$, with every term irrational and the limit rational. For (b), take $r_n = \lfloor 10^n\sqrt{2} \rfloor / 10^n \to \sqrt{2}$, with every term rational and the limit irrational.

7.5 Exercise 7.5: Three differences of large square roots

What we must show. Find the limits of

$$\text{(a) } \sqrt{n^2+1}-n, \qquad \text{(b) } \sqrt{n^2+n}-n, \qquad \text{(c) } \sqrt{4n^2+n}-2n.$$

Plan. Multiply each difference by its conjugate divided by that same conjugate. The difference-of-squares identity removes the subtraction in the numerator. We then bound the error from the proposed limit directly, so no later limit laws are needed.

1. Part (a): Expand the product with the conjugate

Let $a_n = \sqrt{n^2+1}-n$. For $n \geq 1$, the square root is nonnegative and $n > 0$, so the conjugate $\sqrt{n^2+1}+n$ is positive. Dividing by it is valid. Using $(u-v)(u+v) = u^2 - v^2$ gives

$$\begin{aligned} a_n &= \frac{(\sqrt{n^2+1}-n)(\sqrt{n^2+1}+n)}{\sqrt{n^2+1}+n} \\ &= \frac{(\sqrt{n^2+1})^2 - n^2}{\sqrt{n^2+1}+n} \\ &= \frac{n^2+1-n^2}{\sqrt{n^2+1}+n} \\ &= \frac{1}{\sqrt{n^2+1}+n}. \end{aligned}$$

Since $n^2+1 > n^2$ and $n \geq 0$, the nonnegative square root $\sqrt{n^2+1}$ is greater than n. Indeed, the contrary inequality would give $n^2+1 \leq n^2$ upon squaring nonnegative quantities. Therefore

$$0 < a_n = \frac{1}{\sqrt{n^2+1}+n} \leq \frac{1}{2n}.$$

Given $\varepsilon > 0$, choose an integer $N > 1/(2\varepsilon)$. For $n > N$, this estimate gives

$$|a_n - 0| = a_n \leq \frac{1}{2n} < \varepsilon.$$

Thus the limit in part (a) is 0.

2. Part (b): Rationalize and divide by the positive index

Write $b_n = \sqrt{n^2+n}-n$. Its conjugate is positive for $n \geq 1$, and the same multiplication gives

$$\begin{aligned} b_n &= \frac{(\sqrt{n^2+n}-n)(\sqrt{n^2+n}+n)}{\sqrt{n^2+n}+n} \\ &= \frac{(n^2+n)-n^2}{\sqrt{n^2+n}+n} \\ &= \frac{n}{\sqrt{n^2+n}+n}. \end{aligned}$$

The radicand factors as $n^2+n = n^2(1+1/n)$. Because $n > 0$,

$$\sqrt{n^2+n} = n\sqrt{1+\frac{1}{n}}.$$

To check this identity, the right-hand side is nonnegative, and its square is $n^2(1+1/n) = n^2+n$; uniqueness of the nonnegative square root then gives the equality. Define $q_n = \sqrt{1+1/n}$. Substitution and cancellation of $n \neq 0$ yield

$$b_n = \frac{n}{nq_n + n} = \frac{n}{n(q_n+1)} = \frac{1}{q_n+1}.$$

Here $q_n > 1$, since $q_n \geq 0$ and $q_n^2 = 1 + 1/n > 1$; a nonnegative $q_n \leq 1$ would have square at most 1. Hence $q_n + 1 > 2$, so $0 < b_n < 1/2$.

3. Part (b): Compute the exact error from $1/2$

All the denominators below are positive. Since $b_n < 1/2$,

$$\begin{aligned}
\left| b_n - \frac{1}{2} \right| &= \frac{1}{2} - \frac{1}{q_n+1} \\
&= \frac{q_n + 1 - 2}{2(q_n+1)} \\
&= \frac{q_n - 1}{2(q_n+1)} \\
&= \frac{(q_n-1)(q_n+1)}{2(q_n+1)^2} \\
&= \frac{q_n^2 - 1}{2(q_n+1)^2} \\
&= \frac{(1+1/n) - 1}{2(q_n+1)^2} \\
&= \frac{1}{2n(q_n+1)^2} \leq \frac{1}{8n}.
\end{aligned}$$

The last inequality uses $(q_n+1)^2 \geq 2^2 = 4$ and $n > 0$. For any $\varepsilon > 0$, choose an integer $N > 1/(8\varepsilon)$. Then $n > N$ implies $|b_n - 1/2| \leq 1/(8n) < \varepsilon$. Therefore the limit in part (b) is $1/2$.

4. Part (c): Rationalize and factor the square root

Put $c_n = \sqrt{4n^2+n} - 2n$. Since $n \geq 1$, its conjugate $\sqrt{4n^2+n} + 2n$ is positive. Consequently

$$\begin{aligned}
c_n &= \frac{(\sqrt{4n^2+n} - 2n)(\sqrt{4n^2+n} + 2n)}{\sqrt{4n^2+n} + 2n} \\
&= \frac{(4n^2+n) - (2n)^2}{\sqrt{4n^2+n} + 2n} \\
&= \frac{4n^2 + n - 4n^2}{\sqrt{4n^2+n} + 2n} \\
&= \frac{n}{\sqrt{4n^2+n} + 2n}.
\end{aligned}$$

Now $4n^2 + n = n^2(4 + 1/n)$. The positive-index square-root identity gives

$$\sqrt{4n^2+n} = n\sqrt{4 + \frac{1}{n}},$$

because both sides are nonnegative and the square of the right-hand side is $n^2(4 + 1/n) = 4n^2 + n$. Let $p_n = \sqrt{4 + 1/n}$. Canceling the nonzero factor n gives

$$c_n = \frac{n}{np_n + 2n} = \frac{n}{n(p_n + 2)} = \frac{1}{p_n + 2}.$$

Since $p_n \geq 0$ and $p_n^2 = 4 + 1/n > 4$, we have $p_n > 2$. Thus $p_n + 2 > 4$ and $0 < c_n < 1/4$.

5. Part (c): Compute the exact error from $1/4$

Using $c_n < 1/4$ and then multiplying numerator and denominator by the positive number $p_n + 2$, we obtain

$$\begin{aligned}
\left| c_n - \frac{1}{4} \right| &= \frac{1}{4} - \frac{1}{p_n + 2} \\
&= \frac{p_n + 2 - 4}{4(p_n + 2)} \\
&= \frac{p_n - 2}{4(p_n + 2)} \\
&= \frac{(p_n - 2)(p_n + 2)}{4(p_n + 2)^2} \\
&= \frac{p_n^2 - 4}{4(p_n + 2)^2} \\
&= \frac{(4 + 1/n) - 4}{4(p_n + 2)^2} \\
&= \frac{1}{4n(p_n + 2)^2} \leq \frac{1}{64n}.
\end{aligned}$$

Here $(p_n + 2)^2 \geq 4^2 = 16$, so the denominator is at least $4n \cdot 16 = 64n$. For any $\varepsilon > 0$, choose an integer $N > 1/(64\varepsilon)$. Then $n > N$ gives $|c_n - 1/4| \leq 1/(64n) < \varepsilon$. This proves that the limit in part (c) is $1/4$.

Conclusion. The limits are (a) 0, (b) 1/2, and (c) 1/4. The displayed conjugate identities and error estimates justify all three values directly.

8 A discussion about proofs

Every limit proof in this section starts with an arbitrary $\varepsilon > 0$, chooses an appropriate threshold, and verifies the required inequality for every integer index beyond that threshold. An exploratory rearrangement helps discover N; the formal argument then runs forward from $n > N$. Divergence proofs instead show that the definition cannot hold for any proposed real limit. No limit law from Section 9 is assumed.

A proof template for the first exercises

Definition 7.1 says that $x_n \to L$ means

for every real $\varepsilon > 0$ there exists a number N such that every integer $n > N$ satisfies $|x_n - L| < \varepsilon$.

The order of these choices matters: the reader supplies an arbitrary positive ε; we choose N in terms of that ε; then we verify the estimate for *every* sequence index $n > N$. Our choice of N cannot depend on which later index is tested.

Ross permits a real threshold N. If an integer threshold is preferred, the Archimedean property supplies a natural number larger than our chosen N, and the same proof applies to all indices above that larger threshold. The first finitely many terms need not obey the error estimate: the definition requires it only after the threshold.

All the limit assertions below are proved from this definition. Algebraic simplifications, sign rules, and absolute-value identities are used as rules about individual real numbers, without invoking the limit theorems of Section 9.

8.1 Exercise 8.1: Four limits proved directly from the definition

What we must show. Prove each of the four stated limits with a complete ε–N argument.

Part (a): the alternating reciprocal sequence.

We must prove

$$\lim_{n\to\infty} \frac{(-1)^n}{n} = 0.$$

For an integer n, $(-1)^n$ is either 1 or -1, according to the parity of n. Its absolute value is therefore always 1. Since $n \in \mathbb{N}$ implies $n > 0$, the exact error is

$$\begin{aligned}\left|\frac{(-1)^n}{n} - 0\right| &= \frac{|(-1)^n|}{n}, \\ &= \frac{1}{n}.\end{aligned}$$

We need this error to be strictly less than ε.

Let $\varepsilon > 0$ be arbitrary and choose

$$N = \max\left\{1, \frac{1}{\varepsilon}\right\}.$$

Take any integer $n > N$. Then $n > 1/\varepsilon$. Multiplication by the positive number ε gives $\varepsilon n > 1$; division by the positive number n gives $\varepsilon > 1/n$. Consequently

$$\left|\frac{(-1)^n}{n} - 0\right| = \frac{1}{n} < \varepsilon.$$

The estimate holds for every integer $n > N$, as the definition requires.

Conclusion. For every $\varepsilon > 0$ the displayed threshold works, so $\boxed{(-1)^n/n \longrightarrow 0}$. Alternation does not interfere with the estimate because both possible signs have absolute value 1.

Part (b), Step 1: compute the error and the needed threshold.

We must prove $1/n^{1/3} \to 0$, where $n^{1/3}$ is the positive real cube root of n. Since $n^{1/3} > 0$,

$$\left|\frac{1}{n^{1/3}} - 0\right| = \frac{1}{n^{1/3}}.$$

To make this less than ε, it suffices to have $n^{1/3} > 1/\varepsilon$, or $n > 1/\varepsilon^3$. These implications concern ordinary positive real numbers. For example, strict comparison of their cubes follows from

$$u^3 - v^3 = (u - v)(u^2 + uv + v^2) \qquad (u, v > 0).$$

If $u > v > 0$, both factors on the right are positive, so $u^3 > v^3$. Conversely, if $u^3 > v^3$ but $u \le v$, cubing the nonnegative numbers would give $u^3 \le v^3$, a contradiction. Thus comparing positive cubes is equivalent to comparing their positive cube roots.

Part (b), Step 2: give the formal proof.

Let $\varepsilon > 0$ be arbitrary and choose

$$N = \max\left\{1, \frac{1}{\varepsilon^3}\right\}.$$

For an arbitrary integer $n > N$, we have

$$n > \frac{1}{\varepsilon^3} = \left(\frac{1}{\varepsilon}\right)^3.$$

Taking positive cube roots gives $n^{1/3} > 1/\varepsilon$. Multiplying this inequality by $\varepsilon > 0$ and then dividing by $n^{1/3} > 0$ gives

$$\varepsilon n^{1/3} > 1 \quad \Longrightarrow \quad \frac{1}{n^{1/3}} < \varepsilon.$$

Hence the exact error satisfies $\left|1/n^{1/3} - 0\right| < \varepsilon$ for every integer $n > N$.

Conclusion. Definition 7.1 therefore gives $\boxed{1/n^{1/3} \longrightarrow 0}$. The threshold depends on ε^{-3} because the denominator is a cube root.

Part (c), Step 1: simplify the exact difference from the proposed limit.

The sequence in the printed exercise is

$$x_n = \frac{2n - 1}{3n + 2}, \qquad \text{and the proposed limit is } \frac{2}{3}.$$

For every $n \in \mathbb{N}$, $3n+2 > 0$, so the denominator is nonzero. Putting the two fractions over the common denominator $3(3n+2)$ gives

$$\begin{aligned} x_n - \frac{2}{3} &= \frac{3(2n-1) - 2(3n+2)}{3(3n+2)}, \\ &= \frac{6n - 3 - 6n - 4}{9n+6}, \\ &= \frac{-7}{9n+6}. \end{aligned}$$

Because $9n+6 > 0$, taking absolute values yields

$$\left| x_n - \frac{2}{3} \right| = \frac{7}{9n+6}.$$

Also $9n+6 \geq 9n > 0$, so increasing the positive denominator decreases the quotient:

$$\frac{7}{9n+6} \leq \frac{7}{9n}.$$

Part (c), Step 2: choose a threshold and check all later indices.

Let $\varepsilon > 0$ and set

$$N = \max\left\{1, \frac{7}{9\varepsilon}\right\}.$$

If an integer $n > N$, then $n > 7/(9\varepsilon)$. Multiplication by $9\varepsilon > 0$ gives $9\varepsilon n > 7$; division by $9n > 0$ gives $\varepsilon > 7/(9n)$. Thus

$$\left| x_n - \frac{2}{3} \right| = \frac{7}{9n+6} \leq \frac{7}{9n} < \varepsilon.$$

Conclusion. The estimate holds for every integer $n > N$, proving

$$\boxed{\lim_{n\to\infty} \frac{2n-1}{3n+2} = \frac{2}{3}}.$$

Part (d), Step 1: isolate a positive tail of the denominator.

We must prove

$$x_n = \frac{n+6}{n^2-6} \longrightarrow 0.$$

The denominator is nonzero for every positive integer n: if $n \leq 2$, then $n^2 \leq 4 < 6$; if $n \geq 3$, then $n^2 \geq 9 > 6$. For the limit estimate, we will use only indices $n \geq 4$. For such n, we have $n^2/2 \geq 8 \geq 6$, and therefore

$$n^2 - 6 \geq n^2 - \frac{n^2}{2} = \frac{n^2}{2} > 0.$$

This is a lower bound on the denominator, with its sign recorded.

Part (d), Step 2: bound the numerator and then the entire error.

For $n \geq 1$, the inequality $6 \leq 6n$ gives

$$n + 6 \leq n + 6n = 7n.$$

If $n \geq 4$, both the numerator $n+6$ and denominator n^2-6 are positive. Thus

$$\begin{aligned} |x_n - 0| &= \frac{n+6}{n^2-6}, \\ &\leq \frac{7n}{n^2-6} && \text{by the upper bound on the numerator,} \\ &\leq \frac{7n}{n^2/2} && \text{by the positive lower bound on the denominator,} \\ &= \frac{14n}{n^2} = \frac{14}{n}. \end{aligned}$$

The cancellation in the last line is valid because $n > 0$.

Part (d), Step 3: give the formal threshold.

Let $\varepsilon > 0$ and choose

$$N = \max\left\{4, \frac{14}{\varepsilon}\right\}.$$

An arbitrary integer $n > N$ satisfies $n > 4$, so every preceding tail estimate applies. It also satisfies $n > 14/\varepsilon$; multiplying by $\varepsilon > 0$ and dividing by $n > 0$ gives $14/n < \varepsilon$. Hence

$$\left|\frac{n+6}{n^2-6} - 0\right| \leq \frac{14}{n} < \varepsilon \qquad \text{for every integer } n > N.$$

Conclusion. The threshold supplies the estimate required by Definition 7.1, so

$$\boxed{\frac{n+6}{n^2-6} \longrightarrow 0}.$$

We did not require the first few terms to have a positive denominator.

8.2 Exercise 8.2: Determine five limits and justify each one formally

What we must show. Find the limits of the five stated sequences and prove every answer directly from Definition 7.1. The candidate for a quotient of linear expressions can be suggested by its leading coefficients, but the error calculations below are what establish the limits.

Part (a), Step 1: estimate the difference from zero.

For $a_n = n/(n^2+1)$, the numerator and denominator are positive for every $n \in \mathbb{N}$. Since $n^2 + 1 \geq n^2 > 0$,

$$\begin{aligned} |a_n - 0| &= \frac{n}{n^2+1}, \\ &\leq \frac{n}{n^2}, \\ &= \frac{1}{n}. \end{aligned}$$

We will prove that 0 is the limit.

Part (a), Step 2: supply the quantified proof.

Let $\varepsilon > 0$ and set

$$N = \max\left\{1, \frac{1}{\varepsilon}\right\}.$$

For every integer $n > N$, we have $n > 1/\varepsilon$. Because ε and n are positive, this implies $1/n < \varepsilon$. The error bound from Step 1 then gives

$$|a_n - 0| \leq \frac{1}{n} < \varepsilon.$$

Conclusion. The limit is $\boxed{\lim_{n\to\infty} \frac{n}{n^2+1} = 0}$, as verified for every positive error tolerance.

Part (b), Step 1: compute the exact difference from $7/3$.

Let $b_n = (7n-19)/(3n+7)$. Since $n \geq 1$, its denominator satisfies $3n+7 \geq 10 > 0$. Compute

$$\begin{aligned} b_n - \frac{7}{3} &= \frac{3(7n-19) - 7(3n+7)}{3(3n+7)}, \\ &= \frac{21n - 57 - 21n - 49}{9n+21}, \\ &= \frac{-106}{9n+21}. \end{aligned}$$

Thus

$$\left|b_n - \frac{7}{3}\right| = \frac{106}{9n+21} \leq \frac{106}{9n},$$

because $9n + 21 \geq 9n > 0$.

Part (b), Step 2: choose and verify a threshold.

Given any $\varepsilon > 0$, choose

$$N = \max\left\{1, \frac{106}{9\varepsilon}\right\}.$$

Take any integer $n > N$. Then $n > 106/(9\varepsilon)$, so $9\varepsilon n > 106$ and hence $106/(9n) < \varepsilon$. It follows that

$$\left|b_n - \frac{7}{3}\right| = \frac{106}{9n+21} \leq \frac{106}{9n} < \varepsilon.$$

Conclusion. Since this works for every $\varepsilon > 0$,

$$\boxed{\lim_{n\to\infty} \frac{7n-19}{3n+7} = \frac{7}{3}}.$$

Part (c), Step 1: compute the exact difference from $4/7$.

Let $c_n = (4n+3)/(7n-5)$. Its denominator is positive for all natural indices, since $7n-5 \geq 7-5 = 2 > 0$. Cross-multiplication gives

$$\begin{aligned} c_n - \frac{4}{7} &= \frac{7(4n+3) - 4(7n-5)}{7(7n-5)}, \\ &= \frac{28n + 21 - 28n + 20}{49n - 35}, \end{aligned}$$

$$= \frac{41}{49n - 35}.$$

The error is already positive, so its absolute value has the same expression.

Part (c), Step 2: bound the denominator below by a multiple of n.

For $n \geq 1$,

$$(7n - 5) - 2n = 5n - 5 = 5(n - 1) \geq 0.$$

Consequently $7n - 5 \geq 2n > 0$, and multiplying by $7 > 0$ gives $49n - 35 \geq 14n > 0$. Hence

$$\left| c_n - \frac{4}{7} \right| = \frac{41}{49n - 35} \leq \frac{41}{14n}.$$

Part (c), Step 3: prove the definition with an explicit threshold.

Let $\varepsilon > 0$ and put

$$N = \max\left\{1, \frac{41}{14\varepsilon}\right\}.$$

For an arbitrary integer $n > N$, the inequality $n > 41/(14\varepsilon)$ gives $14\varepsilon n > 41$ and thus $41/(14n) < \varepsilon$. Therefore

$$\left| c_n - \frac{4}{7} \right| \leq \frac{41}{14n} < \varepsilon.$$

The estimates used here hold for every such index, not merely for a selected collection of large indices.

Conclusion. The limit is

$$\boxed{\lim_{n\to\infty} \frac{4n + 3}{7n - 5} = \frac{4}{7}}.$$

Part (d), Step 1: simplify the difference from $2/5$.

For $d_n = (2n + 4)/(5n + 2)$, the denominator satisfies $5n + 2 \geq 7 > 0$. The exact calculation is

$$\begin{aligned} d_n - \frac{2}{5} &= \frac{5(2n + 4) - 2(5n + 2)}{5(5n + 2)}, \\ &= \frac{10n + 20 - 10n - 4}{25n + 10}, \\ &= \frac{16}{25n + 10}. \end{aligned}$$

Thus

$$\left| d_n - \frac{2}{5} \right| = \frac{16}{25n + 10} \leq \frac{16}{25n},$$

where the comparison uses $25n + 10 \geq 25n > 0$.

Part (d), Step 2: choose a threshold and verify the required error.

For an arbitrary $\varepsilon > 0$, set

$$N = \max\left\{1, \frac{16}{25\varepsilon}\right\}.$$

If an integer $n > N$, then $n > 16/(25\varepsilon)$. Multiplication by $25\varepsilon > 0$ gives $25\varepsilon n > 16$; division by $25n > 0$ gives $16/(25n) < \varepsilon$. Hence

$$\left| d_n - \frac{2}{5} \right| \leq \frac{16}{25n} < \varepsilon.$$

Conclusion. Definition 7.1 gives

$$\boxed{\lim_{n\to\infty} \frac{2n+4}{5n+2} = \frac{2}{5}}.$$

Part (e), Step 1: bound the error using the elementary sine bound.

The sequence is $s_n = (\sin n)/n$. For every real x, the elementary trigonometric inequality $-1 \leq \sin x \leq 1$ gives $|\sin x| \leq 1$. Applying it at $x = n$ and using $n > 0$ yields

$$\begin{aligned} |s_n - 0| &= \left|\frac{\sin n}{n}\right|, \\ &= \frac{|\sin n|}{n}, \\ &\leq \frac{1}{n}. \end{aligned}$$

The bound holds for every natural n regardless of the sign of $\sin n$.

Part (e), Step 2: carry out the formal proof.

Let $\varepsilon > 0$ and choose

$$N = \max\left\{1, \frac{1}{\varepsilon}\right\}.$$

For every integer $n > N$, multiplication of $n > 1/\varepsilon$ by $\varepsilon > 0$ and division by $n > 0$ give $1/n < \varepsilon$. Consequently

$$|s_n - 0| \leq \frac{1}{n} < \varepsilon.$$

Conclusion. Thus $\boxed{(\sin n)/n \longrightarrow 0}$. The proof uses a uniform bound on sine, not a claim that the sequence $(\sin n)$ has a limit.

8.3 Exercise 8.3: Taking square roots when the limit is zero

What we must show. Suppose $s_n \geq 0$ for every $n \in \mathbb{N}$ and $s_n \to 0$. Prove that $\sqrt{s_n} \to 0$ directly from Definition 7.1.

Plan. To make the nonnegative square root smaller than ε, it suffices to make the original number smaller than ε^2. The input error tolerance for the given convergence will therefore be $\delta = \varepsilon^2$.

Step 1: choose the input tolerance and use the hypothesis.

Let $\varepsilon > 0$ be arbitrary. Then

$$\delta = \varepsilon^2 > 0.$$

The assertion $s_n \to 0$ means that the definition applies to every positive tolerance, including this particular δ. Hence there is an N such that every integer $n > N$ satisfies

$$|s_n - 0| < \delta = \varepsilon^2.$$

Because $s_n \geq 0$, its absolute value is itself, and the last estimate is exactly

$$0 \leq s_n < \varepsilon^2.$$

Step 2: justify the strict comparison of square roots.

Fix any integer $n > N$ and put $u = \sqrt{s_n}$. The nonnegative square root exists because $s_n \geq 0$, and it satisfies $u \geq 0$ and $u^2 = s_n$. If $u \geq \varepsilon$, then

$$u^2 - \varepsilon^2 = (u - \varepsilon)(u + \varepsilon) \geq 0,$$

since $u - \varepsilon \geq 0$ and $u + \varepsilon > 0$. This would imply $s_n = u^2 \geq \varepsilon^2$, contradicting the estimate from Step 1. Thus $u < \varepsilon$.

Step 3: verify the target absolute-value estimate.

Since $u \geq 0$,

$$|\sqrt{s_n} - 0| = \sqrt{s_n} = u < \varepsilon \qquad \text{for every integer } n > N.$$

The threshold N was supplied by the given convergence for the tolerance ε^2; it does not depend on the later index n.

Conclusion. For every target tolerance $\varepsilon > 0$ we obtained the required threshold, proving $\boxed{\sqrt{s_n} \longrightarrow 0}$. No division by $\sqrt{s_n}$ was used, so the argument includes terms equal to zero.

8.4 Exercise 8.4: A sequence tending to zero times a bounded sequence

What we must show. Assume (t_n) is bounded, so there is a real number M with $|t_n| \leq M$ for every $n \in \mathbb{N}$, and assume $s_n \to 0$. Prove $s_n t_n \to 0$ directly from the definition.

Step 1: choose a positive bound that can be used as a divisor.

The given bound satisfies $M \geq |t_1| \geq 0$, so $M \geq 0$. It need not be strictly positive. Define instead

$$B = \max\{1, M\}.$$

Then $B \geq 1 > 0$, and for every n,

$$|t_n| \leq M \leq B.$$

The constant B is fixed for the whole sequence; it does not depend on the index n or on our later choice of error tolerance.

Step 2: ask the sequence tending to zero for a sufficiently small error.

Let $\varepsilon > 0$ be arbitrary and put

$$\delta = \frac{\varepsilon}{B} > 0.$$

Since $s_n \to 0$, there is an N such that, for every integer $n > N$,

$$|s_n - 0| < \delta = \frac{\varepsilon}{B}.$$

Step 3: verify the product estimate term by term.

Take an arbitrary integer $n > N$. Multiplication of $|t_n| \leq B$ by the nonnegative number $|s_n|$ gives $|s_n|\,|t_n| \leq B\,|s_n|$. Multiplication of $|s_n| < \varepsilon/B$ by the strictly positive number B preserves the strict inequality. Therefore

$$|s_n t_n - 0| = |s_n|\,|t_n|\,,$$

$$\begin{aligned} &\leq B\,|s_n|\,, \\ &< B\frac{\varepsilon}{B}, \\ &= \varepsilon. \end{aligned}$$

This bound holds for every integer $n > N$.

Step 4: check the possible zero-bound case.

If the original M was zero, then $0 \leq |t_n| \leq 0$ forces $t_n = 0$ for every n. The products are then all zero. Our proof already covers this case because it used $B = \max\{1, 0\} = 1$ rather than dividing by $M = 0$.

Conclusion. The error estimate establishes $\boxed{s_n t_n \longrightarrow 0}$ by Definition 7.1. The sequence (t_n) need only be bounded; its own convergence was not assumed or used.

8.5 Exercise 8.5: The squeeze lemma and an absolute-value comparison

What we must show. Prove (a) that $a_n \leq s_n \leq b_n$ for all n, together with $a_n \to s$ and $b_n \to s$, implies $s_n \to s$; and (b) that $|s_n| \leq t_n$ for all n, together with $t_n \to 0$, implies $s_n \to 0$.

Part (a), Step 1: obtain two thresholds for the same tolerance.

Let $\varepsilon > 0$ be arbitrary. Because $a_n \to s$, there is an N_1 such that

$$n > N_1 \quad \Longrightarrow \quad |a_n - s| < \varepsilon.$$

Because $b_n \to s$, there is an N_2 such that

$$n > N_2 \quad \Longrightarrow \quad |b_n - s| < \varepsilon.$$

Each implication is required for every integer index exceeding the corresponding threshold.

Part (a), Step 2: make both estimates valid at once.

Choose

$$N = \max\{N_1, N_2\}.$$

For an arbitrary integer $n > N$, we have both $n > N_1$ and $n > N_2$. Thus both error estimates apply at this same index. The elementary absolute-value equivalence from Exercise 3.7(a) gives

$$-\varepsilon < a_n - s < \varepsilon, \qquad -\varepsilon < b_n - s < \varepsilon.$$

Part (a), Step 3: insert the middle term between these errors.

The order assumption holds for every n, so in particular for our chosen $n > N$ we have $a_n \leq s_n \leq b_n$. Add $-s$ throughout:

$$a_n - s \leq s_n - s \leq b_n - s.$$

Combine the strict lower bound from a_n with the strict upper bound from b_n:

$$-\varepsilon < a_n - s \leq s_n - s \leq b_n - s < \varepsilon.$$

The middle inequalities are permitted to be weak. The outer strict inequalities still give $-\varepsilon < s_n - s < \varepsilon$ by transitivity. Applying Exercise 3.7(a) once more gives $|s_n - s| < \varepsilon$.

Part (a), Step 4: record the quantifiers of the conclusion.

The number $\varepsilon > 0$ was arbitrary. We chose a single N for that tolerance and verified $|s_n - s| < \varepsilon$ for every integer $n > N$. This is exactly the definition of convergence to s.

Conclusion. The squeeze lemma is

$$\boxed{a_n \leq s_n \leq b_n, \quad a_n \to s, \quad b_n \to s \quad \Longrightarrow \quad s_n \to s}.$$

The two enclosing sequences may approach s from either side; only their common limit and the displayed order comparisons were used.

Part (b), Step 1: extract the sign information contained in the hypothesis.

For each n, $|s_n| \geq 0$. The assumed inequality therefore gives

$$0 \leq |s_n| \leq t_n.$$

In particular, $t_n \geq 0$ for every index. This is a consequence of the hypothesis, not an extra assumption.

Part (b), Step 2: apply the definition to the given sequence.

Let $\varepsilon > 0$ be arbitrary. Since $t_n \to 0$, there is an N such that every integer $n > N$ satisfies

$$|t_n - 0| < \varepsilon.$$

As $t_n \geq 0$, this is the same as $t_n < \varepsilon$.

Part (b), Step 3: finish the required error estimate.

For every integer $n > N$, combine Step 1 with Step 2:

$$0 \leq |s_n - 0| = |s_n| \leq t_n < \varepsilon.$$

Thus $|s_n - 0| < \varepsilon$ for every such n.

Conclusion. For every $\varepsilon > 0$ the preceding threshold works, so $\boxed{s_n \longrightarrow 0}$. This direct proof records why the sign of t_n causes no difficulty.

8.6 Exercise 8.6: Absolute values at zero and an alternating counterexample

What we must show. Prove (a) $s_n \to 0$ if and only if $|s_n| \to 0$, and (b) the sequence $s_n = (-1)^n$ has convergent absolute values but is itself divergent.

Throughout these exercises, convergence is used in the precise form of Definition 7.1: $x_n \to L$ means that for every $\varepsilon > 0$ there is a real number N such that

$$n \in \mathbb{N}, \quad n > N \quad \Longrightarrow \quad |x_n - L| < \varepsilon.$$

A threshold N need not be an integer. Whenever an index above a specified threshold is needed, the Archimedean property supplies a positive integer above it.

Part (a), Step 1: compare the two errors exactly.

For every $n \in \mathbb{N}$, the number $|s_n|$ is nonnegative. Therefore

$$||s_n| - 0| = ||s_n|| = |s_n| = |s_n - 0|.$$

The errors in the two proposed zero limits are the same real number, not merely bounded by each other.

Part (a), Step 2: prove the forward implication.

Assume $s_n \to 0$, and let $\varepsilon > 0$ be arbitrary. The definition supplies N such that

$$n > N \quad \Longrightarrow \quad |s_n - 0| < \varepsilon.$$

For each such index, the identity in Step 1 gives

$$||s_n| - 0| = |s_n - 0| < \varepsilon.$$

Thus the same threshold N verifies convergence of $|s_n|$ to 0.

Part (a), Step 3: prove the reverse implication.

Assume $|s_n| \to 0$, and let $\varepsilon > 0$ be arbitrary. Choose N from this convergence so that

$$n > N \quad \Longrightarrow \quad ||s_n| - 0| < \varepsilon.$$

Using Step 1 again yields

$$|s_n - 0| = ||s_n| - 0| < \varepsilon \qquad (n > N).$$

This is the definition of $s_n \to 0$.

Conclusion. Both implications have been proved with the same error and the same threshold:

$$\boxed{s_n \to 0 \quad \Longleftrightarrow \quad |s_n| \to 0.}$$

Part (b), Step 1: prove convergence of the absolute values.

If $s_n = (-1)^n$, then s_n equals either 1 or -1. Hence $|s_n| = 1$ for every n. For any $\varepsilon > 0$, choose $N = 0$. Every positive integer $n > N$ satisfies

$$||s_n| - 1| = |1 - 1| = 0 < \varepsilon.$$

Therefore $|s_n| \to 1$.

Part (b), Step 2: rule out every possible limit of the original sequence.

Suppose, for a contradiction, that $s_n \to L$ for some real L. Apply the definition with the fixed positive tolerance $\varepsilon = 1$. There is a real N such that

$$n > N \quad \Longrightarrow \quad |s_n - L| < 1.$$

Choose an integer $k > \max\{N, 1\}$. Then both $2k$ and $2k + 1$ are positive integers larger than N. Their terms are

$$s_{2k} = 1, \qquad s_{2k+1} = -1.$$

The triangle inequality gives

$$\begin{aligned} 2 &= |1 - (-1)| \\ &= |(1 - L) + (L - (-1))| \end{aligned}$$

$$\begin{aligned}
&\leq |1 - L| + |L - (-1)| \\
&= |s_{2k} - L| + |s_{2k+1} - L| \\
&< 1 + 1 = 2.
\end{aligned}$$

The equality of the absolute values in the fourth line uses $|-x| = |x|$. The resulting strict inequality $2 < 2$ is impossible.

Conclusion. The absolute-value sequence converges to 1, but $((-1)^n)$ has no real limit. Thus existence of a limit for $|s_n|$ alone does not guarantee existence of a limit for s_n.

8.7 Exercise 8.7: Three divergent sequences

What we must show. Show that none of the following sequences converges:

$$\cos\left(\frac{n\pi}{3}\right), \qquad (-1)^n n, \qquad \sin\left(\frac{n\pi}{3}\right).$$

Plan. We will use only the definition of convergence and the triangle inequality. If a sequence converged to L, then for any chosen $\varepsilon > 0$, all sufficiently late terms would be within ε of L. Any two such terms would satisfy

$$|s_p - s_q| = |(s_p - L) + (L - s_q)| \leq |s_p - L| + |s_q - L| < 2\varepsilon.$$

For each proposed sequence, we will produce two arbitrarily late terms whose distance contradicts this inequality. This displayed observation has just been proved directly, so no theorem about subsequences or bounded convergent sequences is needed.

Part (a), Step 1: select a tolerance under an assumed limit.

Suppose

$$s_n = \cos\left(\frac{n\pi}{3}\right) \to L.$$

With $\varepsilon = 1$, choose N such that every positive integer $n > N$ satisfies $|s_n - L| < 1$.

Part (a), Step 2: find late terms equal to 1 and -1.

Choose an integer $k > \max\{N, 1\}$, and put

$$p = 6k, \qquad q = 6k + 3.$$

Both indices exceed N. The period 2π of cosine gives

$$s_p = \cos(2k\pi) = 1, \qquad s_q = \cos(2k\pi + \pi) = -1.$$

Their distance is therefore $|s_p - s_q| = 2$. The two-late-terms estimate gives

$$2 = |s_p - s_q| \leq |s_p - L| + |s_q - L| < 1 + 1 = 2,$$

a contradiction.

Conclusion. The sequence $(\cos(n\pi/3))$ does not converge.

Part (b), Step 1: use the definition with a fixed tolerance.

Suppose $s_n = (-1)^n n \to L$. Choose N from the definition with $\varepsilon = 1$.

Part (b), Step 2: compute the distance between late even and odd terms.

Choose an integer $k > \max\{N, 1\}$. Then $p = 2k$ and $q = 2k + 1$ are larger than N and

$$s_p = 2k, \qquad s_q = -(2k + 1).$$

Their difference is

$$s_p - s_q = 2k + (2k + 1) = 4k + 1 > 2.$$

Since this difference is positive, its absolute value is also $4k + 1$. But convergence would give

$$4k + 1 = |s_p - s_q| \leq |s_p - L| + |s_q - L| < 1 + 1 = 2.$$

This contradicts $4k + 1 > 2$.

Conclusion. The sequence $((-1)^n n)$ does not converge. The proof uses two explicit terms in every sufficiently late tail, rather than assuming a boundedness theorem from a later section.

Part (c), Step 1: choose a tolerance suited to the sine values.

Suppose

$$s_n = \sin\left(\frac{n\pi}{3}\right) \to L.$$

Choose the positive tolerance $\varepsilon = \sqrt{3}/2$. The definition gives N such that $n > N$ implies $|s_n - L| < \sqrt{3}/2$.

Part (c), Step 2: use two positions in the six-term period.

Choose an integer $k > \max\{N, 1\}$ and let

$$p = 6k + 1, \qquad q = 6k + 4.$$

Both are larger than N. By the period 2π of sine,

$$s_p = \sin\left(2k\pi + \frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}, \qquad s_q = \sin\left(2k\pi + \frac{4\pi}{3}\right) = -\frac{\sqrt{3}}{2}.$$

In particular, $|s_p - s_q| = \sqrt{3}$. The convergence estimates would imply

$$\sqrt{3} = |s_p - s_q| \leq |s_p - L| + |s_q - L| < \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} = \sqrt{3}.$$

This is impossible.

Conclusion. The sequence $(\sin(n\pi/3))$ does not converge. Every sequence in the exercise has now been ruled out by the definition of convergence itself.

8.8 Exercise 8.8: Three radical limits with explicit error bounds

What we must show. Prove the three stated limits:

$$\sqrt{n^2 + 1} - n \longrightarrow 0,$$

$$\sqrt{n^2+n}-n \longrightarrow \frac{1}{2},$$
$$\sqrt{4n^2+n}-2n \longrightarrow \frac{1}{4}.$$

The index n is always a positive integer. In particular, division by n is legitimate, and $\sqrt{n^2}=n$ rather than $-n$.

Plan. Multiplication by a conjugate removes the subtraction of large radicals. We will then bound each *error from its claimed limit* by a positive multiple of $1/n$. The estimates below are ordinary algebraic identities and inequalities, not applications of limit laws.

Part (a), Step 1: rationalize the expression and check its sign.

Let $u_n=\sqrt{n^2+1}-n$. Because $n\geq 1$, both n and $\sqrt{n^2+1}$ are positive. Moreover, $\sqrt{n^2+1}>n$: otherwise squaring a supposed inequality $\sqrt{n^2+1}\leq n$ between nonnegative numbers would give $n^2+1\leq n^2$, a contradiction. Thus $u_n>0$. Its conjugate $\sqrt{n^2+1}+n$ is positive, so

$$\begin{aligned} u_n &= \frac{(\sqrt{n^2+1}-n)(\sqrt{n^2+1}+n)}{\sqrt{n^2+1}+n} \\ &= \frac{(n^2+1)-n^2}{\sqrt{n^2+1}+n} \\ &= \frac{1}{\sqrt{n^2+1}+n}. \end{aligned}$$

Part (a), Step 2: obtain an error estimate valid for every index.

The denominator satisfies $\sqrt{n^2+1}+n\geq 2n>0$. Taking reciprocals of positive quantities reverses the comparison, so

$$|u_n-0|=u_n\leq \frac{1}{2n}.$$

Part (a), Step 3: choose and verify a threshold.

Let $\varepsilon>0$ be arbitrary, and choose

$$N=\frac{1}{2\varepsilon}.$$

For every positive integer $n>N$, multiplying by $2\varepsilon>0$ gives $2\varepsilon n>1$. Division by $2n>0$ gives $1/(2n)<\varepsilon$. Hence

$$n>N \quad\Longrightarrow\quad |u_n-0|\leq \frac{1}{2n}<\varepsilon.$$

This is precisely $u_n\to 0$.

Conclusion. $\boxed{\lim_{n\to\infty}(\sqrt{n^2+1}-n)=0}$. An explicit error bound is $1/(2n)$.

Part (b), Step 1: rationalize and factor out the positive index.

Let $v_n=\sqrt{n^2+n}-n$. Multiplying by its positive conjugate gives

$$v_n=\frac{(n^2+n)-n^2}{\sqrt{n^2+n}+n}=\frac{n}{\sqrt{n^2+n}+n}.$$

Set

$$r_n = \sqrt{1 + \frac{1}{n}}.$$

The quantity nr_n is positive, and its square is

$$(nr_n)^2 = n^2\left(1 + \frac{1}{n}\right) = n^2 + n.$$

By uniqueness of the nonnegative square root, $\sqrt{n^2+n} = nr_n$. Substitution gives the exact identity

$$v_n = \frac{n}{nr_n + n} = \frac{1}{r_n + 1}.$$

The number r_n is greater than 1, because $r_n^2 = 1 + 1/n > 1$. Thus $r_n + 1 > 2$, and $0 < v_n < 1/2$.

Part (b), Step 2: rationalize the error from $1/2$.

The sign information in Step 1 gives

$$\begin{aligned}\left|v_n - \tfrac{1}{2}\right| &= \frac{1}{2} - \frac{1}{r_n + 1} \\ &= \frac{(r_n + 1) - 2}{2(r_n + 1)} \\ &= \frac{r_n - 1}{2(r_n + 1)}.\end{aligned}$$

A second conjugate calculation yields

$$r_n - 1 = \frac{(r_n - 1)(r_n + 1)}{r_n + 1} = \frac{r_n^2 - 1}{r_n + 1} = \frac{1/n}{r_n + 1}.$$

Therefore

$$\left|v_n - \tfrac{1}{2}\right| = \frac{1}{2n(r_n + 1)^2}.$$

Since $r_n \geq 1$, we have $(r_n + 1)^2 \geq 4$, so

$$\left|v_n - \tfrac{1}{2}\right| \leq \frac{1}{8n}.$$

Part (b), Step 3: verify the definition of convergence.

Let $\varepsilon > 0$ be arbitrary and choose $N = 1/(8\varepsilon)$. For every positive integer $n > N$, multiplication and division by positive quantities give $1/(8n) < \varepsilon$. Consequently

$$\left|v_n - \tfrac{1}{2}\right| \leq \frac{1}{8n} < \varepsilon \qquad (n > N).$$

The definition proves the desired limit.

Conclusion. $\boxed{\lim_{n\to\infty}(\sqrt{n^2+n} - n) = \tfrac{1}{2}}$. An explicit error bound is $1/(8n)$.

Part (c), Step 1: rationalize and identify a simpler denominator.

Let $w_n = \sqrt{4n^2+n} - 2n$. Its conjugate is positive, and multiplication gives

$$w_n = \frac{(4n^2+n) - 4n^2}{\sqrt{4n^2+n} + 2n} = \frac{n}{\sqrt{4n^2+n} + 2n}.$$

Define $R_n = \sqrt{4 + 1/n}$. Since $n > 0$, the quantity nR_n is the nonnegative square root of $4n^2 + n$:

$$(nR_n)^2 = n^2\left(4 + \frac{1}{n}\right) = 4n^2 + n.$$

Hence

$$w_n = \frac{n}{nR_n + 2n} = \frac{1}{R_n + 2}.$$

Because $R_n^2 = 4 + 1/n > 4$ and $R_n \geq 0$, we have $R_n > 2$. Thus $R_n + 2 > 4$ and $0 < w_n < 1/4$.

Part (c), Step 2: compute and bound the error from $1/4$.

Using the sign of the error, put the fractions over one denominator:

$$\begin{aligned}\left|w_n - \tfrac{1}{4}\right| &= \frac{1}{4} - \frac{1}{R_n + 2} \\ &= \frac{(R_n + 2) - 4}{4(R_n + 2)} \\ &= \frac{R_n - 2}{4(R_n + 2)}.\end{aligned}$$

The second rationalization gives

$$R_n - 2 = \frac{R_n^2 - 4}{R_n + 2} = \frac{1/n}{R_n + 2}.$$

It follows that

$$\left|w_n - \tfrac{1}{4}\right| = \frac{1}{4n(R_n + 2)^2} \leq \frac{1}{64n},$$

because $R_n \geq 2$ implies $(R_n + 2)^2 \geq 16$.

Part (c), Step 3: give the final tolerance-dependent threshold.

Let $\varepsilon > 0$ be arbitrary and choose $N = 1/(64\varepsilon)$. Then $n > N$ implies $1/(64n) < \varepsilon$, so

$$\left|w_n - \tfrac{1}{4}\right| \leq \frac{1}{64n} < \varepsilon \qquad (n \in \mathbb{N},\ n > N).$$

This verifies convergence directly.

Conclusion. $\boxed{\lim_{n\to\infty}(\sqrt{4n^2+n} - 2n) = \tfrac{1}{4}}$. All three limits were proved by explicit inequalities, without using algebraic limit theorems from Section 9.

8.9 Exercise 8.9: Eventual bounds pass to a convergent limit

What we must show. Let $s_n \to s$. Prove:

(a) If $s_n \geq a$ for all but finitely many indices, then $s \geq a$.

(b) If $s_n \le b$ for all but finitely many indices, then $s \le b$.

(c) If $s_n \in [a, b]$ for all but finitely many indices, then $s \in [a, b]$.

A preliminary detail: replace finitely many exceptions by one threshold.

If a property fails only on a finite set $E \subseteq \mathbb{N}$ of indices, define $K = 0$ if E is empty and $K = \max E$ otherwise. Every positive integer $n > K$ is outside E, so every such index satisfies the property. This explains each use of an eventual-bound threshold below.

Part (a), Step 1: suppose the limit violates the eventual lower bound.

Choose K so that $n > K$ implies $s_n \ge a$. Suppose, for a contradiction, that $s < a$. Then the gap $a - s$ is positive. Set

$$\varepsilon = \frac{a - s}{2} > 0.$$

Convergence supplies N_0 such that $n > N_0$ implies $|s_n - s| < \varepsilon$.

Part (a), Step 2: combine the two tail conditions.

Choose a positive integer $n > \max\{K, N_0\}$. The condition $n > K$ gives $s_n \ge a$. The condition $n > N_0$ gives

$$s_n - s < \varepsilon, \qquad s_n < s + \varepsilon = \frac{s + a}{2}.$$

But

$$a - \frac{s + a}{2} = \frac{a - s}{2} > 0.$$

Thus $s_n < (s + a)/2 < a$, contradicting $s_n \ge a$.

Conclusion. The inequality $s < a$ is impossible. Therefore $\boxed{s \ge a}$. Equality $s = a$ is permitted; for example, the constant sequence $s_n = a$ has limit a and satisfies every lower bound in the hypothesis.

Part (b), Step 1: suppose the limit violates the eventual upper bound.

Choose K so that $n > K$ implies $s_n \le b$. Suppose $s > b$, and choose the positive tolerance

$$\varepsilon = \frac{s - b}{2}.$$

Convergence gives N_0 such that $n > N_0$ implies $|s_n - s| < \varepsilon$.

Part (b), Step 2: choose one index after both thresholds.

Take a positive integer $n > \max\{K, N_0\}$. Then $s_n \le b$, whereas the convergence inequality gives

$$-\varepsilon < s_n - s, \qquad s_n > s - \varepsilon = \frac{s + b}{2}.$$

The midpoint exceeds b because

$$\frac{s + b}{2} - b = \frac{s - b}{2} > 0.$$

Hence $s_n > b$, contradicting $s_n \le b$.

Conclusion. The inequality $s > b$ is impossible. Therefore $\boxed{s \le b}$. The endpoint $s = b$ is allowed.

Part (c), Step 1: unpack membership of a closed interval.

For $a \le b$,

$$x \in [a, b] \quad \Longleftrightarrow \quad a \le x \le b.$$

If all but finitely many s_n belong to this interval, then all but finitely many satisfy $s_n \ge a$, and all but finitely many satisfy $s_n \le b$. The finite-exception threshold may be chosen from the indices where interval membership fails, so both inequalities hold beyond the same threshold.

Part (c), Step 2: apply both proved statements to the same limit.

Part (a) gives $s \ge a$, and part (b) gives $s \le b$. Together these say $a \le s \le b$, which is exactly $s \in [a, b]$. If $a = b$, the two inequalities force $s = a = b$, so the singleton case is included as well.

Conclusion. For a convergent sequence,

$$\boxed{s_n \in [a, b] \text{ for all but finitely many } n \quad \Longrightarrow \quad \lim s_n \in [a, b].}$$

The finitely many exceptional early terms do not affect any of these conclusions because an index can be chosen after every exception and after the convergence threshold.

8.10 Exercise 8.10: A strict bound for the limit becomes an eventual strict bound

What we must show. If $s_n \to s$ and $s > a$, prove that there is a real N such that

$$n \in \mathbb{N}, \quad n > N \quad \Longrightarrow \quad s_n > a.$$

Plan. The positive gap $s - a$ allows us to place a whole ε-neighborhood of the limit above a.

Step 1: choose a tolerance smaller than the gap.

Because $s > a$, the real number $s - a$ is positive. Define

$$\varepsilon = \frac{s - a}{2} > 0.$$

The tolerance is fixed from the limit and the bound; it does not depend on the index n.

Step 2: obtain a threshold from convergence.

The definition of $s_n \to s$ gives a real number N such that

$$n \in \mathbb{N}, \quad n > N \quad \Longrightarrow \quad |s_n - s| < \varepsilon.$$

For any index in this tail, the absolute-value inequality implies

$$-\varepsilon < s_n - s < \varepsilon.$$

The left inequality gives $s_n > s - \varepsilon$.

Step 3: check that the lower edge is above the desired bound.

Substitution and simplification give

$$s - \varepsilon = s - \frac{s - a}{2} = \frac{2s - s + a}{2} = \frac{s + a}{2}.$$

Also

$$\frac{s+a}{2} - a = \frac{s-a}{2} > 0,$$

so $(s+a)/2 > a$. Combining these strict comparisons yields

$$s_n > s - \varepsilon = \frac{s+a}{2} > a \qquad (n \in \mathbb{N},\ n > N).$$

Conclusion. The threshold supplied for $\varepsilon = (s-a)/2$ satisfies the required implication:

$$\boxed{n > N \quad \Longrightarrow \quad s_n > a.}$$

The strict hypothesis $s > a$ matters: if $s = a$ and $s_n = a$ for all n, the sequence converges to a but none of its terms is greater than a.